

Question Paper

Exam Date & Time: 26-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: CSE 3121/CSE_3121 PRINCIPLES OF CRYPTOGRAPHY

(SPL: COMPUTER SCIENCE & ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Evaluate the output of the Static S box in Advanced Encryption Standard (AES-128) encryption algorithm for the input 0x3C (Show all the iterations with all the polynomial long divisions steps, and the other necessary steps) (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1B) With a neat diagram, explain the symmetric cipher model. The encryption process used was a combination of Columnar transposition Cipher (key=PRINCIPLE) and then followed by Rail Fence Cipher (depth 3) to obtain the ciphertext "TYRCRORYPETPPHAOPG". Decrypt to get plaintext. (Assuming padding with 'X' is used if required and only alphabets (case-insensitive) are considered). (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Analyze and apply Euler's theorem to determine the value of $7^{364} \bmod 120$ showing each intermediate computational step. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2A) With a neat diagram, evaluate how cryptographic hash functions ensure message integrity and authenticity. Analyze how a slight change in bits affects the output. Justify why this property guarantees tamper detection in integrity verification systems. (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 2B) Inspect if there exists a period while generating 10 random numbers using Linear Congruential Generator where modulus=1000, multiplier=21, increment=30 and seed=7. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) Mention any 4 key features of Data Encryption Standard (DES) algorithm. Explain the working of DES encryption using a neat diagram. (3)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 3A) A student claims that the unique solution x for the following set of congruences is $x=23$ (4)

$$x \equiv 2 \pmod{5}$$

$$x \equiv 3 \pmod{6}$$

$$x \equiv 2 \pmod{7}$$

Evaluate the validity of this claim using the Chinese Remainder Theorem.
If the claim is incorrect, determine the correct value of x and explain where the reasoning went wrong.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 3B) A plaintext (PT) of two characters is encrypted using its ASCII equivalents, with each character encrypted separately under RSA. The key generation primes are $p = 13$ and $q = 23$, and the public exponent is $e = 37$. The ciphertext (CT) sequence is (66,134). Evaluate the process by calculating the private key and decrypting the ciphertext to recover the plaintext message. Show the modular exponential steps. (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3C) Explain the working of Shift Rows Transformation of **AES-128** encryption algorithm on an Input state and with the help of an example, explain the advantage of applying the transformation. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 4A) Given a single column of an AES state matrix: (3)

$$s = \begin{pmatrix} 97 \\ EC \\ C3 \\ 95 \end{pmatrix}$$

Compute the MixColumns transformation for this column using the AES fixed matrix. Explain how mix columns enhance AES security

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 4B) Using Diffie-Hellman key exchange process, two parties, Alice and Bob, wish to securely exchange a shared secret key over an insecure channel without revealing the key to an eavesdropper. For the given values like prime number(q)=7, primitive root (α) =5, Alice Private Key (X_a)=6 and Bob Private Key (X_b)=7. Calculate Alice and Bob's public keys Y_a and Y_b . Explain the process of exchanging public keys between Alice and Bob. Compute the shared secret key that both Alice and Bob. Verify that Alice and Bob arrive at the same shared secret key. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Analyse counter (CTR) mode and output feedback (OFB) mode of Pseudo Random Number Generation Mechanism Based on Block Ciphers? (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) Summarize various types of functions that may be used to produce an authenticator. Explain basic Uses of Message Encryption with a neat diagram? (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) Suppose a cryptanalyst encounters a set of blocks encoded with the RSA algorithm, but the private key is unknown. Given that $n=pq$ and e is the public key, how would the cryptanalyst know that one of the plaintext blocks shares a common factor with n which assists in determining the private key? (4)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5C) Illustrate with a neat diagram various ways through which a hash code can be used to provide digital signature? (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_3121:Principles of Cryptography

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	13
3	3	CO 3	12
4	4	CO 4	12
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_3121:Principles of Cryptography

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	4	13(26%)
4	Analysis	6	21(42%)
5	Create	4	14(28%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)